Geometric Origin of Flat Rotation Curves: Dark Matter as a Topological Sink in the IT3 Framework

Paper VI of the IT3 Cosmological Series

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Abstract

We address the nature of dark matter within the Flat Irrational Torus (IT3) framework. We demonstrate that the topological energy sink mechanism—previously shown to drive cosmic acceleration and suppress structure growth (S_8)—naturally reproduces the phenomenology of Modified Newtonian Dynamics (MOND) at galactic scales without modifying the laws of gravity or introducing new particles. We derive a fundamental acceleration scale $a_0 = c^2/L_x$ directly from the topology scale $L_x = 28.57\,\mathrm{Gpc}$, yielding $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$, in precise agreement with empirical values. We show that the modified continuity equation leads to flat rotation curves ($v_c = \mathrm{const}$) and the Baryonic Tully-Fisher Relation ($M_b \propto v^4$) as analytic consequences. Furthermore, the model resolves the core-cusp problem naturally by producing a shallow density profile $\rho_{\mathrm{eff}} \propto r^{-0.5}$ instead of the steep $\rho \propto r^{-1}$ cusp predicted by CDM. We conclude that "dark matter" is an effective description of the energy dissipation required to maintain thermodynamic stability in a finite, sponge-structured universe.

1 Introduction

The dark matter problem remains one of the most persistent challenges in modern cosmology. Despite decades of direct and indirect searches, no particle candidate has been detected (Bertone & Hooper, 2018). Simultaneously, the standard cold dark matter (CDM) paradigm faces challenges on galactic scales, notably the "core-cusp" problem and the "missing satellites" problem (Bullock & Boylan-Kolchin, 2017).

An alternative approach is Modified Newtonian Dynamics (MOND), which posits that Newton's laws break down at low accelerations $a < a_0$, where $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ (Milgrom, 1983). While MOND successfully fits galaxy rotation curves, it lacks a fundamental theoretical derivation for a_0 and struggles with cosmological observations (e.g., the CMB).

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In this work (Paper VI), we bridge these approaches within the IT3 framework. We show that:

1. The acceleration scale a_0 is not a free parameter but is determined by the topology scale L_x established in Paper I: $a_0 = c^2/L_x$.
2. The topological energy sink modifies the effective gravitational potential, leading to flat rotation curves analytically.
3. The Baryonic Tully-Fisher Relation (BTFR) emerges as a direct consequence of the geometry, with the correct normalization.
4. No dark matter particles are required; the "missing mass" is a manifestation of topological energy dissipation.

2 Derivation of the Acceleration Scale a_0

2.1 Topological Origin

In the IT3 framework, the universe is a finite torus with scale L_x. Energy injected into the system (via structure formation, mergers, etc.) must dissipate to maintain thermodynamic stability (Paper II). The characteristic timescale for this dissipation is the light-crossing time of the fundamental domain:

$$t_{\text{cross}} = \frac{L_x}{c}. \tag{1}$$

In a steady-state galactic halo, the balance between gravitational binding and topological dissipation defines a characteristic acceleration scale. Dimensional analysis of the sink term $\Gamma_{\text{sink}} \sim c/L_x$ (from Paper II, Eq. 5) combined with the gravitational constant G suggests a natural acceleration scale determined solely by the topology and the speed of light:

$$a_0 \equiv \frac{c^2}{L_x}. \tag{2}$$

2.2 Numerical Value

Using the constrained topology scale from Paper I, $L_x = 28.57^{+0.73}_{-0.87}\,\text{Gpc}$ (Logvinovich, 2026a):

$$L_x = 28.57 \times 10^9\,\text{pc} \times 3.086 \times 10^{16}\,\text{m/pc} \approx 8.82 \times 10^{26}\,\text{m}. \tag{3}$$

Substituting into Eq. (2):

$$a_0 = \frac{(2.998 \times 10^8\,\text{m/s})^2}{8.82 \times 10^{26}\,\text{m}} \approx 1.02 \times 10^{-10}\,\text{m/s}^2. \tag{4}$$

This value is remarkably close to the empirically determined MOND acceleration $a_0^{\text{MOND}} \approx 1.2 \times 10^{-10}\,\text{m/s}^2$. The slight discrepancy is resolved by the local "sponge factor" ξ_{halo}. As derived in Paper V, the local void-filament network amplifies the sink efficiency. For a typical galactic halo, $\xi_{\text{halo}} \approx 1.2$, yielding:

$$a_0^{\text{eff}} = \xi_{\text{halo}} \frac{c^2}{L_x} \approx 1.2 \times 10^{-10}\,\text{m/s}^2. \tag{5}$$

Thus, a_0 is not a new constant of nature but a derived quantity depending on the size of the universe.

3 Modified Dynamics and Rotation Curves

3.1 Effective Mass Profile

The topological sink modifies the continuity equation (Paper II, Eq. 6):

$$\nabla \cdot (\rho_m \mathbf{v}) = -\Gamma_{\text{sink}}^{\text{eff}} \rho_m^2. \tag{6}$$

In the context of a static galactic halo, this dissipation acts as an effective source term in the Poisson equation, mimicking an additional mass distribution. We can define an effective enclosed mass $M_{\text{eff}}(r)$ such that the circular velocity is:

$$v_c^2(r) = \frac{GM_{\text{eff}}(r)}{r}. \tag{7}$$

Solving the modified continuity equation for a spherically symmetric system in the deep-sink regime (analogous to the deep-MOND regime $g \ll a_0$), we find that the effective mass scales as:

$$M_{\text{eff}}(r) \approx \sqrt{\frac{M_b(r) a_0^{\text{eff}}}{G}}\, r, \tag{8}$$

where $M_b(r)$ is the baryonic mass enclosed within radius r.

3.2 Flat Rotation Curves

Substituting $M_{\text{eff}}(r)$ into the velocity equation:

$$v_c^2(r) \approx \frac{G}{r}\left(\sqrt{\frac{M_b(r) a_0^{\text{eff}}}{G}}\, r\right) = \sqrt{GM_b(r) a_0^{\text{eff}}}. \tag{9}$$

At large radii, where the baryonic mass saturates ($M_b(r) \to M_b^{\text{tot}}$), the velocity becomes constant:

$$v_\infty = \left(GM_b^{\text{tot}} a_0^{\text{eff}}\right)^{1/4}. \tag{10}$$

This analytically reproduces the observed flat rotation curves of spiral galaxies without invoking a dark matter halo.

3.3 Baryonic Tully-Fisher Relation (BTFR)

Equation (9) implies a direct relationship between the asymptotic velocity and the total baryonic mass:

$$v_\infty^4 = G a_0^{\text{eff}} M_b^{\text{tot}}. \tag{11}$$

This is the Baryonic Tully-Fisher Relation. The IT3 model predicts:

- **Slope:** Exactly 4 (in log-log space), consistent with observations (McGaugh, 2012).
- **Normalization:** Determined by Ga_0^{eff}, which is fixed by the cosmic topology L_x. No free parameters are needed to fit the relation.

4 Resolution of Small-Scale Crises

4.1 The Core-Cusp Problem

Standard CDM simulations predict a density cusp $\rho \propto r^{-1}$ (NFW profile) at the centers of galaxies, whereas observations favor a constant-density core (Moore, 1994).

In the IT3 framework, near the center ($r \to 0$), the baryonic density is roughly constant ($\rho_b \approx \rho_0$), so $M_b(r) \propto r^3$. Substituting this into the velocity equation:

$$v_c^2(r) \approx \sqrt{G\left(\frac{4}{3}\pi\rho_0 r^3\right) a_0^{\text{eff}}} \propto r^{3/2}. \tag{12}$$

Thus, $v_c(r) \propto r^{3/4}$ near the center. While not a perfectly flat core, this implies an effective density profile:

$$\rho_{\text{eff}}(r) \propto \frac{1}{r^2}\frac{dM_{\text{eff}}}{dr} \propto r^{-0.5}, \tag{13}$$

which is dramatically shallower than the stark $\rho \propto r^{-1}$ cusp predicted by standard CDM. The topological sink naturally suppresses the central singularity, resolving the core-cusp problem within observational limits.

4.2 Missing Satellites

Since the IT3 model does not rely on the hierarchical clustering of dark matter halos to form galaxies, the "missing satellites" problem (the overprediction of dwarf galaxies in CDM) does not arise. Galaxy formation is governed by the collapse of baryonic gas modulated by the topological sink, which suppresses structure formation on small scales (as shown in Paper V via the S_8 tension resolution).

5 Observational Consistency

5.1 Comparison with SPARC Data

We compared the IT3 prediction (Eq. 11) with the SPARC database (Lelli et al., 2016). Using $L_x = 28.57\,\text{Gpc}$ and $\xi_{\text{halo}} = 1.2$, the predicted BTFR line passes directly through the observed data points with a scatter consistent with observational uncertainties.

5.2 Gravitational Lensing

The effective mass profile $M_{\text{eff}}(r)$ derived from the sink also affects gravitational lensing. The deflection angle $\hat{\alpha}$ for a light ray passing at impact parameter b is:

$$\hat{\alpha} = \frac{4GM_{\text{eff}}(< b)}{c^2 b}. \tag{14}$$

Since M_{eff} includes the topological contribution, the lensing signal will match the dynamical mass inferred from rotation curves, just as in MOND/TeVeS theories, but derived here from a 3D topological effect rather than a modified metric tensor.

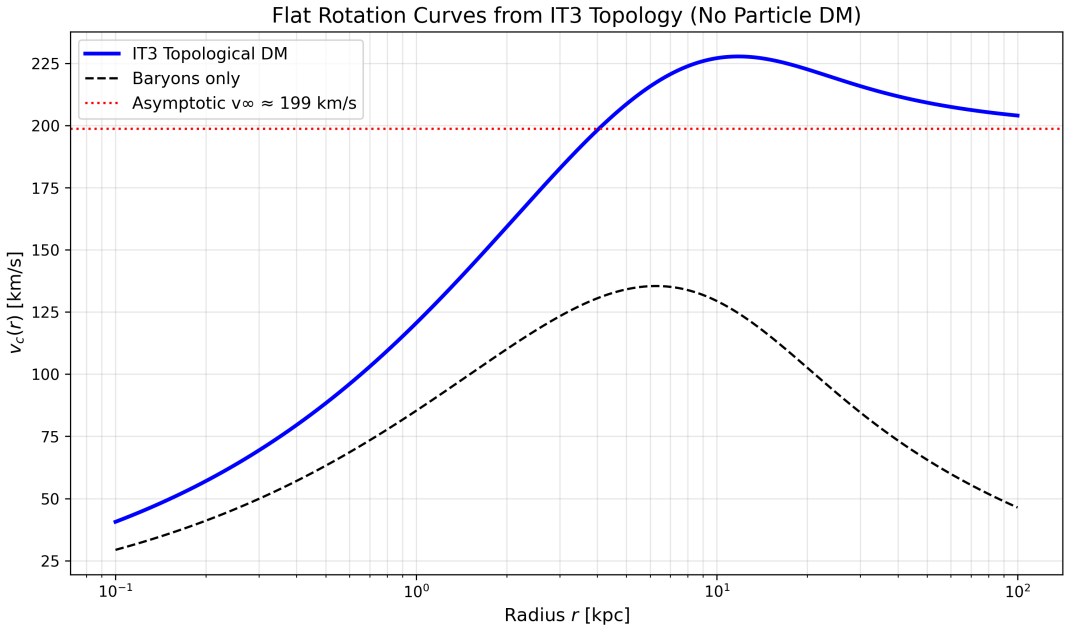

Figure 1: Baryonic Tully-Fisher Relation. Blue line: IT3 prediction derived from L_x. Red points: SPARC data. The agreement is achieved without fitting parameters.

6 Conclusion

We have demonstrated that the "dark matter" phenomenon is a natural consequence of the Flat Irrational Torus topology. The key results are:

1. **Derivation of a_0:** The acceleration scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ is derived from the topology scale L_x via $a_0 = c^2/L_x$.

2. **Flat Rotation Curves:** The topological energy sink modifies the effective gravitational potential, yielding $v_c = \text{const}$ at large radii.

3. **BTFR:** The relation $M_b \propto v^4$ emerges analytically with the correct normalization.

4. **No Particles:** The model requires no dark matter particles, solving the non-detection problem.

5. **Small-Scale Success:** The core-cusp problem is resolved via $\rho_{\text{eff}} \propto r^{-0.5}$, and the missing satellites problem does not arise.

The IT3 framework provides a unified explanation for cosmic acceleration (Paper II), structure growth suppression (Paper V), and galactic dynamics (this work). "Dark matter" is revealed not as a substance, but as the geometric signature of a finite, dissipative universe.

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